

to *slapd.conf* to further protect the rootDN password. Edit */etc/openldap/slapd.conf* and set the encrypted password as follows:

```
[root@vbg ~]# cat /etc/openldap/slapd.conf | grep rootpw
rootpw {SSHA}gX7EhK5K8l14R143r2Q4ME33H0u7fA4
[root@vbg ~]#
```

...and restart the ldap service:

```
[root@vbg ~]# /etc/init.d/ldap restart
Stopping slapd: [ OK ]
config file testing succeeded [ OK ]
Starting slapd: [ OK ]
[root@vbg ~]#
```

How would you test whether the new password is working or not? Do we have to always write something to the directory and check the password?

An easier and risk-free approach to the above would be to search the directory using a user name and password. The syntax for providing the userDN and password in the *ldapsearch* command is similar to that of the *ldapadd* command:

```
[vbg@vbg work ~]$ ldapsearch -x -h 192.168.122.1 -b "dc=ina,dc=org"
# extended LDAP
#
# # LDAPv3
# # base <dc=ina,dc=org> with scope subtree
# # filter: (objectclass=*)
# # requesting: ALL

Anonymous Search works .....
```

Now let us try searching with a user name and an incorrect password:

```
[vbg@vbg work ~]$ ldapsearch -x -h 192.168.122.1 -b "dc=ina,dc=org" -D
"cn=Manager,dc=ina,dc=org" -u xyz
ldap bind: Invalid credentials (49)
```

This fails. Therefore, we know that the supplied credentials (user name/password) are incorrect.

To run a successful search, we will need to specify the correct password. That is left as an exercise for the reader.

Making successful authentication a prerequisite for successful search (disable anonymous directory read) is another security measure that I would recommend. Though this depends on the kind of data maintained in your directory, if you are unsure, it is best to disable anonymous read. This is done through 'access control instructions' in the configuration file.

But before we move on to 'access control restrictions', we first need to understand the structure of *slapd.conf*.

Why did I simply not append a line to the top of the configuration file specifying the encrypted password, above? It would have been very easy to demonstrate by mentioning

the following command, for instance:

```
[root@vbg ~]# MYPASS=$(slappasswd -s tr) ; sed -ie '1i/rootpw "$MYPASS"'
/etc/openldap/slapd.conf
[root@vbg ~]# head -n 1 /etc/openldap/slapd.conf
rootpw {SSHA}L1Hy8VhQ2ndG12ZpUjHrX74wFk3-1TF
[root@vbg ~]#
```

```
[root@vbg ~]# service ldap restart
Stopping slapd: [ OK ]
Checking configuration files for slapd: slaptest: bad configuration file!
[FAILED]
[root@vbg ~]#
```

So, we see that by randomly inserting lines, we are unable to set up the configuration file, *slapd.conf*. This can be achieved in some other configuration files though.

Now, if we delete the first line and add the same line to the bottom of the file, let us see what happens. Delete the first line:

```
[root@vbg ~]# sed -ie '1,10' /etc/openldap/slapd.conf
```

Add the 'rootpw' to the bottom:

```
[root@vbg ~]# echo "rootpw '$MYPASS' $(slappasswd -s POSTER@inglinux)" >> /etc/openldap/slapd.conf
```

This will add the encrypted root password to the configuration file, as demonstrated by the command below:

```
[root@vbg ~]# tail -n 1 /etc/openldap/slapd.conf
rootpw {SSHA}0L8MF7CGHGR2Afa2XNl1on2JfKoiXl
[root@vbg ~]#
```

Now, let us restart the *ldap* service:

```
[root@vbg ~]# service ldap restart
Stopping slapd: [FAILED]
Checking configuration files for slapd: bdb_db_open: Warning - No DB_CONFIG
file found in directory /var/lib/ldap: (2)
Expect poor performance for suffix dc=ina,dc=org.
config file testing succeeded [ OK ]
Starting slapd: [ OK ]
[root@vbg ~]#
```

Obviously, we cannot insert 'rootpw' at the top of *slapd.conf*, but in the case of the default *slapd.conf*, we can do so at the bottom.

Why is it so? This is because *slapd.conf*, follows a certain structure. So let us ask ourselves: what is the need for a structure? One reason that I can think of is, backends.

OpenLDAP backends

Let's take a detailed look at backends in the later part of this series. It is the wide availability of backends that really